Exp: 3 Networks Cables 23/07/2026

AIM:

To study different types of Networking cables.

a)

TYPE	CATEGORY	MAX DATA TRANSMISSION	ADVANTAGES / DISADVANTAGES	APPLICATION
UTP	3	10 bps	Adv: → Cheaper → Easy to Install Dis: → More prone to EMI	10 base-T Ethernet
	5	upto 100 Mbps		Fast Ethernet
	5e	1 Gbps		Gigabit ethernet
STP	6, 6a	10 Gbps	Adv: → shielded → faster than UTP → Less susceptible to noise Dis: → Expensive → Greater Installation effort	Gigabit ethernet 10G Ethernet
SSTP	7	10 Gbps		Gigabit Ethernet
Coaxial cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Adv: → High bandwidth → Versatile Dis: → Limited distance → Cost	Speed of signal is 500 m Television network.
fibre optics cable	Single mode multi mode	100 Gbps	Adv: → High speed → High bandwidth Dis: → Expensive → Requires skilled installers.	Maximum distance of fibre optics cable is arnd 100 meters.

b) Make an Ethernet cross-over cable :

TOOLS :

- → Ethernet cabling
- → A crimping tool
- → Two RJ45 plugs
- → two plug shields

St 1 :

Begin by threading shields onto the cable.

St 2 :

Next, strip approx. 1.5 cm of cable shielding from both ends.

St-3 :

Untangle the wires, thre should be four twisted pairs. One end should be in A & other in B.

St 4 :

Bunch them together in a line, stick out farther than others, snip them back. Hold the plug with clip side facing away from you & pins toward you.

St 5 :

The notch at the end of the plug needs to be just over cable shielding.

St 6 :

After securing the plug, insert into crimping tool

St 7 :

Lastly repeat for other end with B.

Result :

Thus, the study of Network cables (a, b) have been executed & verified successfully.